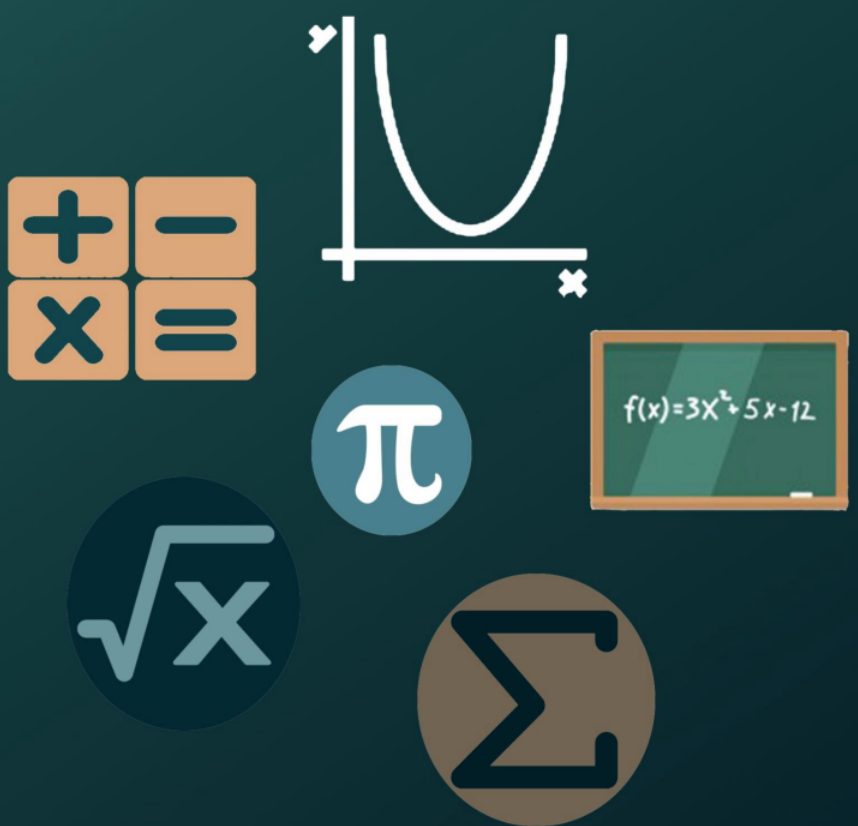
MATHEMATICS

Determinant



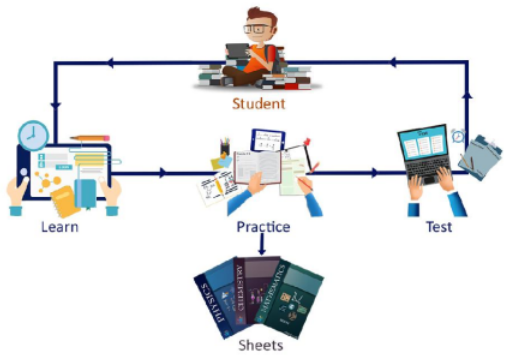
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Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

S.No	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Introduction to Determinants	1,2	1,20	1	10,11,15
2	Minors and Cofactors	3,4	9-12,27,28		
3	Properties of Determinants - Part 1 & 2	5-13	21,22,29		
4	Special Determinants	14-16	2,3,23		
5	Problems on Properties	17-22	4,13-15, 24,30	2-13	1-8,17-22, 26-30
6	Factor Theorem	23-25	31		31
7	Multiplication of Determinants	26,27	5	14,15	23,24
8	Cramer's Rule : System of Linear Equations	28-37	6,7,16-19, 26,32,33	16-20	9,12-17, 32-34
9	Differentiation of Determinants	38-40	8,34	21	16,25

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Determinants

Q1. The value of $\begin{vmatrix} a+1 & a-2 \\ a+2 & a-1 \end{vmatrix}$ is -

(A) $2a^2$ (B) 0

(C) –3 (D) 3

Q2. The value of $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix}$ is-

(A) 213 (B) – 231

(C) 231 (D) 39

Minors and Cofactors

Q3. Find the cofactors of elements '–3', '5', '–1' & '7' in the determinant $\begin{vmatrix} 2 & -3 & 1 \\ 4 & 0 & 5 \\ -1 & 6 & 7 \end{vmatrix}$

(A) (–33, –9, –15, 12)

(B) (33, –9, –15, 12)

(C) (–33, –9, –15, 12)

(D) (33, 9, 15, 12)

Q4. If the value of a third order determinant is 11, then the value of the determinant formed by its cofactors will be

(A) 11 (B) 121

(C) 1331 (D) 14641

Properties of Determinants

Q5. If $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$, then $\begin{vmatrix} ka & kb & kc \\ kx & ky & kz \\ kp & kq & kr \end{vmatrix}$ equals :

(A) $k\Delta$ (B) $3k\Delta$

(C) $k^3\Delta$ (D) $k\Delta^3$

Q6. $\begin{vmatrix} 1 & 5 & \pi \\ \log_e e & 5 & \sqrt{5} \\ \log_{10} 10 & 5 & e \end{vmatrix} =$

(A) $\sqrt{\pi}$ (B) e

(C) 1 (D) 0

Q7. If $\begin{vmatrix} a & b & c \\ m & n & p \\ x & y & z \end{vmatrix} = k$, then $\begin{vmatrix} 6a & 2b & 2c \\ 3m & n & p \\ 3x & y & z \end{vmatrix} =$

(A) $k/6$ (B) $2k$

(C) $3k$ (D) $6k$

Q8. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix} =$

(A) 1 (B) 0

(C) x (D) xy

Q9. $\begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix} =$

(A) $a(x+y+z)+b(p+q+r)+c$

(B) 0

(C) $abc+xyz+pqr$

(D) None of these

Q10. $\begin{vmatrix} 1 & 1 & 1 \\ b+c & c+a & a+b \\ b+c-a & c+a-b & a+b-c \end{vmatrix}$ equals-

(A) 0
(B) ab + bc + ca
(C) abc
(D) a + b + c

Q11. $\begin{vmatrix} b^2+c^2 & a^2 & a^2 \\ b^2 & c^2+a^2 & b^2 \\ c^2 & c^2 & a^2+b^2 \end{vmatrix}$ is equal to-

(A) $a^2b^2c^2$
(B) $2a^2b^2c^2$
(C) $4a^2b^2c^2$
(D) none of these

Q12. If $\begin{vmatrix} 3^2+k & 4^2 & 3^2+3+k \\ 4^2+k & 5^2 & 4^2+4+k \\ 5^2+k & 6^2 & 5^2+5+k \end{vmatrix} = 0$, then the value of k is-

(A) 2
(B) 1
(C) –1
(D) 0

Q13. If α,β,γ are the roots of the equation $x^3+px+q=0$, then the value of the determinant $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ is :

(A) 4
(B) 2
(C) 0
(D) –2

Special Determinants

Q14. $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} =$

(A) $a^2+b^2+c^2$
(B) $(a+b)(b+c)(c+a)$
(C) $(a-b)(b-c)(c-a)$
(D) None of these

Q15. The value of $\begin{vmatrix} 1 & 1 & 1 \\ bc & ca & ab \\ b+c & c+a & a+b \end{vmatrix}$ is -

(A) 1
(B) 0
(C) $(a-b)(b-c)(c-a)$
(D) $(a+b)(b+c)(c+a)$

Q16. If a,b,c are unequal what is the condition that the value of the following determinant is zero $\Delta = \begin{vmatrix} a & a^2 & a^3+1 \\ b & b^2 & b^3+1 \\ c & c^2 & c^3+1 \end{vmatrix}$

(A) $1+abc=0$
(B) $a+b+c+1=0$
(C) $(a-b)(b-c)(c-a)=0$
(D) None of these

Problems on Properties

Q17. If $\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$, then $x =$

(A) 1, 9
(B) – 1, 9
(C) – 1, – 9
(D) 1, – 9

Q18. $\begin{vmatrix} x+1 & x+2 & x+4 \\ x+3 & x+5 & x+8 \\ x+7 & x+10 & x+14 \end{vmatrix} =$

(A) 2
(B) – 2
(C) x^2-2
(D) None of these

Q19. The deteminant $D = \begin{vmatrix} 1 & x & x^2 \\ x^2 & 1 & x \\ x & x^2 & 1 \end{vmatrix}$ is equal to

(A) $\left(1-x^3\right)^2$
(B) $\left(1-x^2\right)^3$
(C) $\left(x-x^2\right)^3$
(D) None of these

Q20. Let a determinant is given by $A=\begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ and suppose det A = 6.

If $B=\begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$ then :

(A) det B = 6 (B) det B = –6
(C) det B = 12 (D) det B = –12

Q21. If $\theta \in (0,\pi/2)$, then value of

$$\begin{vmatrix} (\sin\theta+\operatorname{cosec}\theta)^2 & (\sin\theta-\operatorname{cosec}\theta)^2 & 1 \\ (\cos\theta+\sec\theta)^2 & (\cos\theta-\sec\theta)^2 & 1 \\ (\tan\theta+\cot\theta)^2 & (\tan\theta-\cot\theta)^2 & 1 \end{vmatrix} =$$

(A) $\sin\theta+\cos\theta+\tan\theta$
(B) 1
(C) 0
(D) 4

Q22. If x, y, z are integers in AP, lying between 1 and 9 and x 51, y 41 and z 31 are three digit numbers, then the value of $\begin{vmatrix} 5 & 4 & 3 \\ x51 & y41 & z31 \\ x & y & z \end{vmatrix}$ is -

(A) x + y + z (B) x – y + z
(C) 0 (D) x = 2y + z

Factor Theorem

Q23. If $a \neq b \neq c$, the value of x which satisfies the equation $\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0$, is -

(A) $x = 0$ (B) $x = a$
(C) $x = b$ (D) $x = c$

Q24. $\Delta=\begin{vmatrix} a+x & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix}$,

which of the following is a factor for the given determinant

(A) $x-(a+b+c)$ (B) $x+(a+b+c)$
(C) $a+b+c$ (D) $-(a+b+c)$

Q25. If a,b,c are positive integers, then the determinant $\Delta=\begin{vmatrix} a^2+x & ab & ac \\ ab & b^2+x & bc \\ ac & bc & c^2+x \end{vmatrix}$ is divisible by

(A) x^3 (B) x^2
(C) $(a^2+b^2+c^2)$ (D) None of these

Multiplication of Determinants

Q26. $\begin{vmatrix} \log_3 512 & \log_4 3 \\ \log_3 8 & \log_4 9 \end{vmatrix} \times \begin{vmatrix} \log_2 3 & \log_8 3 \\ \log_3 4 & \log_3 4 \end{vmatrix} =$

(A) 7 (B) 10
(C) 13 (D) 17

Q27. If in the multiplication of $\begin{vmatrix} a & b \\ -b & a \end{vmatrix}$ and $\begin{vmatrix} c & d \\ -d & c \end{vmatrix}$, A and B are the element of the first row then the elements of the second row will be-

(A) –B, A (B) A,B
(C) B, A (D) –B, –A

Cramer's Rule : System of Linear Equations

Q28. The system of equations x + y + z = 2, 2x + y – z = 3, 3x + 2y + kz = 4 has unique solution if-

(A) k = 0 (B) $k \neq 0$
(C) –1 < k < 1 (D) –2 < k < 2

Q29. If a system of the equation
$(\alpha+1)^3 x+(\alpha+2)^3 y-(\alpha+3)^3=0$
and $(\alpha+1)x+(\alpha+2)y-(\alpha+3)=0$,
$x+y-1=0$ is consistent what is the value of α -

(A) 1 (B) 0
(C) -3 (D) -2

Q30. If the lines $ax+by+c=0$, $bx+cy+a=0$ and $cx+ay+b=0$ be concurrent, then

(A) $a^3+b^3+c^3+3abc=0$

(B) $a^3+b^3+c^3-abc=0$

(C) $a^3+b^3+c^3-3abc=0$

(D) None of these

Q31. The following system of equation
$3x-2y+z=0$, $\lambda x-14y+15z=0$,
$x+2y-3z=0$ has a solution other than
$x=y=z=0$ for λ equal to

(A) 1 (B) 2 (C) 3 (D) 5

Q32. If $2x+3y-5z=7, x+y+z=6$,
$3x-4y+2z=1$, then $x=$

(A) $\begin{vmatrix}2&-5&7\\1&1&6\\3&2&1\end{vmatrix} \div \begin{vmatrix}7&3&-5\\6&1&1\\1&-4&2\end{vmatrix}$

(B) $\begin{vmatrix}-7&3&-5\\-6&1&1\\-1&-4&2\end{vmatrix} \div \begin{vmatrix}2&3&-5\\1&1&1\\3&-4&2\end{vmatrix}$

(C) $\begin{vmatrix}7&3&-5\\6&1&1\\1&-4&2\end{vmatrix} \div \begin{vmatrix}2&3&-5\\1&1&1\\3&-4&2\end{vmatrix}$

(D) None of these

Q33. The number of solutions of the equations
$x+4y-z=0$, $3x-4y-z=0$,
$x-3y+z=0$ is

(A) 0 (B) 1
(C) 2 (D) Infinite

Q34. Set of equations $a+b-2c=0$,
$2a-3b+c=0$ and $a-5b+4c=\alpha$ is consistent for α equal to :

(A) 1 (B) 0
(C) -1 (D) 2

Q35. The equations
$\lambda x-y=2, 2x-3y=-\lambda, 3x-2y=-1$
are consistant for :

(A) $\lambda=-4$

(B) $\lambda=-1,4$

(C) $\lambda=-1$

(D) $\lambda=1,-4$

Q36. The number of values of k for which the system of equations.
$(k+1)x+8y=4k$
$kx+(k+3)y=3k-1$ has infinitely many solutions is :

(A) 0
(B) 1
(C) 2
(D) infinite

Q37. The value of λ for which the system of equations $2x-y-z=12$, $x-2y+z=-4$, $x+y+\lambda z=4$ has no solution is

(A) 3 (B) -3
(C) 2 (D) -2

Differentiation of Determinants

Q38. Let $\begin{vmatrix}x&2&x\\x^2&x&6\\x&x&6\end{vmatrix}=Ax^4+Bx^3+cx^2+Dx+E$,

then the value of $5A+4B+3C+2D+E$ is equal to:

(A) -17 (B) -14
(C) -11 (D) -8

Q39. If $\Delta(x)=\begin{vmatrix} x^n & \sin x & \cos x \\ n! & \sin\frac{n\pi}{2} & \cos\frac{n\pi}{2} \\ a & a^2 & a^3 \end{vmatrix}$, then the value of $\frac{d^n}{dx^n}[\Delta(x)]$ at $x=0$ is

(A) -1 (B) 0

(C) 1 (D) Dependent of a

Q40. Let $f(x)=\begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$ where p is a constant. Then $\frac{d^3(f(x))}{dx^3}$ at $x=0$ is

(A) $6p^3$ (B) $p+p^2$

(C) $p+p^3$ (D) independent of p

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}=0$, then choose INCORRECT option(s) from the following (where a,b,c are distinct positive real numbers)

(A) a,b,c must be AP (B) a,b,c must be GP

(C) a,b,c must be HP (D) a,b,c can't be AP

Q2. Which of the following determinant(s) reduces to zero-

(A) $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix}$

(B) $\begin{vmatrix} 1 & 2 & 3 \\ 51 & 52 & 53 \\ 201 & 202 & 203 \end{vmatrix}$

(C) $\begin{vmatrix} 0 & a-b & b-c \\ b-a & 0 & c-a \\ c-b & a-c & 0 \end{vmatrix}$

(D) $\begin{vmatrix} a & \lambda a+\mu b & b \\ p & \lambda p+\mu q & q \\ u & \lambda u+\mu\omega & \omega \end{vmatrix}$

Q3. If $\Delta=\begin{vmatrix} 1 & (a+b) & (a+b)^2 \\ 1 & (b+c) & (b+c)^2 \\ 1 & (c+a) & (c+a)^2 \end{vmatrix}$, where a,b,c $\in$ distinct positive integer, then-

(A) minimum value of $\Delta=2$

(B) minimum value of $\Delta=1$

(C) Δ may be a prime number

(D) Δ is always an even integer

Q4. $f(x)$ is a polynomial defined as $f(x)=\begin{vmatrix} x^2 & 4 & 4 \\ 4 & x^2 & 4 \\ 4 & 4 & x^2 \end{vmatrix}$, then $f(x)$ is divisible by

(A) x^2+4+4x (B) x^2+4-4x

(C) x^2+8 (D) x^2-8

Q5. If $D=\begin{vmatrix} 2 & 1 & 1 \\ x^2+y^2 & x^2+y^2 & 2xy \\ 2xy & 2xy & x^2+y^2 \end{vmatrix}$

and $D_1=\begin{vmatrix} 1 & 0 & 0 \\ 0 & x & y \\ 0 & y & x \end{vmatrix}$,

then determinant D_2, such that $D_2=\frac{D}{D_1}$ is-

(A) $(x+y)(x-y)$ (B) $(x^2-y^2)^2$

(C) $\begin{vmatrix} 2 & 1 & 1 \\ y & y & x \\ x & x & y \end{vmatrix}$ (D) $\begin{vmatrix} 2 & x & y \\ 1 & x & y \\ 1 & y & x \end{vmatrix}$

Q6. If the system of equations $x+2y+z=px$, $-3x+py=2z$ & $2x-2y+z=-pz$ has non-trivial solution then value of p may be equal to -

(A) 1 (B) 0

(C) –3 (D) 3

Q7. Consider the system of equations

$(\lambda+1)x+y+\lambda z=3$, $\lambda x+2\lambda y+z=4$,

$3x+(\lambda-1)y+\lambda z=2$, $\lambda \in R$

then identify the correct statement/s-

(A) there are exactly three values of λ for which system has unique solution

(B) there are infinite values of λ for which system has unique solution

(C) there is no value of λ for which system has unique solution

(D) there is no value of λ for which system has exactly two solutions

Q8. If $f(x)=\begin{vmatrix} \sin x & \cos x & 1 \\ \cos x & \sin x & 1 \\ 1 & \sin x & \cos x \end{vmatrix}$ then

identify the correct statement(s) -

(A) Number of solutions of $f(x)=0$ is six in $[0,2\pi]$

(B) Number of solutions of $f(x)=0$ is three in $[0,2\pi]$

(C) $f'(0)=1$

(D) $f'(0)=0$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 9 to 12)

Consider the determinant $\Delta=\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

M_{ij} = Minor of the element of ith row & j^{th} column.

C_{ij} = Cofactor of element of ith row & j^{th} column.

Q9. Value of $b_1.C_{31}+b_2.C_{32}+b_3.C_{33}$ is :-

(A) 0 (B) Δ

(C) 2Δ (D) Δ^2

Q10. $a_2.C_{12}+b_2.C_{22}+c_2.C_{32}$ is equal to :-

(A) 0 (B) Δ

(C) 2Δ (D) Δ^2

Q11. If all the elements of the determinant are multiplied by 2, then the value of new determinant is :-

(A) 0 (B) 8Δ

(C) 2Δ (D) $2^9\Delta$

Q12. $a_3M_{13}-b_3M_{23}+c_3M_{33}$ is equal to :-

(A) 0 (B) 4Δ

(C) 2Δ (D) Δ

Paragraph # 2 (Ques. 13 to 15)

Let $x, y, z \in R^+$ & $D = \begin{vmatrix} x & x^3 & x^4-1 \\ y & y^3 & y^4-1 \\ z & z^3 & z^4-1 \end{vmatrix}$

Q13. If $x \neq y \neq z$ & x, y, z are in GP and D = 0, then y is equal to -

(A) 1

(B) 2

(C) 4

(D) none of these

Q14. If x, y, z are the roots of $t^3 - 21t^2 + bt - 343 = 0$, $b \in R$, then D is equal to-

(A) 1

(B) 0

(C) dependent on x, y, z

(D) data inadequate

Q15. If $x \neq y \neq z$ & x, y, z are in A.P. and D = 0, then $2xy^2z + x^2z^2$ is equal to-

(A) 1

(B) 2

(C) 3

(D) none of these

Paragraph # 3 (Ques.16 to 18)

Consider the system of linear equations

$\alpha x + y + z = m$

$x + \alpha y + z = n$

and $x + y + \alpha z = p$

Q16. If $\alpha \neq 1, -2$ then the system has -

(A) no solution

(B) infinte solutions

(C) unique solution

(D) trivial solution if $m \neq n \neq p$

Q17. If $\alpha = -2$ & $m + n + p \neq 0$ then system of linear equations has -

(A) no solution

(B) infinite solutions

(C) unique solution

(D) finitely many solution

Q18. If $\alpha = 1$ & $m \neq p$ then the system of linear equations has -

(A) no solution

(B) infinite solutions

(C) unique solution

(D) unique solution if p = n

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q19.

Column-I	**Column-II**
(A) If the determinant $\begin{vmatrix} a+p & \ell+x & u+f \\ b+q & m+y & v+g \\ c+r & n+z & w+h \end{vmatrix}$ splits into exactly K determinants of order 3, each element of which contains only one term, then the values of K is	(P) 3

(B) The values of λ for which the system of equations
$x + y + z = 6$,
$x + 2y + 3z = 10$
& $x + 2y + \lambda z = 12$
is inconsistent — (Q) 8

(C) If x, y, z are in A.P. then the value of the determinant
$\begin{vmatrix} a+2 & a+3 & a+2x \\ a+3 & a+4 & a+2y \\ a+4 & a+5 & a+2z \end{vmatrix}$ is — (R) 5

(D) Let p be the sum of all possible determinants of order 2 having 0, 1, 2 & 3 as their four elements (without repeatition of digits). The value of 'p' is — (S) 0

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q20. Let $f(x) = \begin{vmatrix} x & 1 & \frac{-3}{2} \\ 2 & 2 & 1 \\ \frac{1}{x-1} & 0 & \frac{1}{2} \end{vmatrix}$.

Find the minimum value of $f(x)$ (given $x > 1$).

Q21. If a, b and c are pth, qth and rth terms of an H.P. then $\begin{vmatrix} bc & ca & ab \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix} =$

Q22. Find the value of the determinant
$\begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix}$

Q23. If $D_r = \begin{vmatrix} r & r^3 & 2 \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix}$,
find $\sum_{r=0}^{n} D_r$

Q24. If
$f(x) = \begin{vmatrix} 6-\cos^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 5+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 5+4\sin 2x \end{vmatrix}$,
where $x \in R$.
Find maximum value of $f(x)$.

Q25. If $\begin{vmatrix} \sin 3\theta & -1 & 1 \\ \cos 2\theta & 4 & 3 \\ 2 & 7 & 7 \end{vmatrix} = 0$, then find the number of values of θ in $[0, 2\pi]$.

Q26. If the system of equations $\lambda x + (b-a)y + (c-a)z = 0$, $(a-b)x + \lambda y + (c-b)z = 0$ and $(a-c)x + (b-c)y + \lambda z = 0$ has a non-trivial solution, then the value of λ is

SUBJECTIVE TYPE

Q27. Find the minor and cofactor of 6 in the determinant $\begin{vmatrix} 5 & 6 & 1 \\ 2 & 3 & -5 \\ 4 & 0 & 7 \end{vmatrix}$

Q28. Write the cofactor of each element in the determinant $\begin{vmatrix} a & b \\ -c & d \end{vmatrix}$

Q29. Prove that $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

Q30. Prove that $\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix} = -(1-\alpha^3)^2$

Q31. Prove that $\begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = m(x-a)(x-b)$

Q32. Find the nature of solution for the given system of equations :
$x + 2y + 3z = 1;\ \ 2x + 3y + 4z = 3;$
$3x + 4y + 5z = 0$

Q33. If the system of equations $x + \lambda y + 1 = 0$, $\lambda x + y + 1 = 0$ & $x + y + \lambda = 0$. is consistent, then find the value of λ.

Q34. If $y = \begin{vmatrix} x^2 & 2x \\ 5 & x^3+2 \end{vmatrix}$ then find $\frac{dy}{dx}$

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Let $\begin{vmatrix} \tan^{-1}x & \tan^{-1}2x & \tan^{-1}3x \\ \tan^{-1}3x & \tan^{-1}x & \tan^{-1}2x \\ \tan^{-1}2x & \tan^{-1}3x & \tan^{-1}x \end{vmatrix} = 0$

then number of values of x satisfying the equation is :

(A) 1 (B) 2
(C) 3 (D) 4

Q2. The determinant

$$\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0 \text{ if}$$

(A) x, y, z are in A.P.
(B) x,y, z are in G.P.
(C) x, y, z are in H.P.
(D) xy, yz, zx are in A.P.

Q3. If $D_p = \begin{vmatrix} p & 15 & 8 \\ p^2 & 35 & 9 \\ p^3 & 25 & 10 \end{vmatrix}$, then

$D_1 + D_2 + D_3 + D_4 + D_5 =$

(A) 0 (B) 25
(C) 625 (D) None of these

Q4. The value of $\sum_{n=1}^{N} U_n$, if

$$U_n = \begin{vmatrix} n & 1 & 5 \\ n^2 & 2N+1 & 2N+1 \\ n^3 & 3N^2 & 3N \end{vmatrix}$$ is

(A) 0 (B) 1

(C) – 1 (D) None of these

Q5. When is $\begin{vmatrix} y+z & z & y \\ z & z+x & x \\ y & x & x+y \end{vmatrix}$ divided by xyz, the quotient will be

(A) 4xyz (B) 2xyz

(C) 4 (D) 2

Q6. If $\sum_{n=1}^{n} \alpha_n = pn^2 + qn + r$ and

$\alpha_1, \alpha_2, \alpha_3 \in \{1, 2, 3, ...9\}$ and

$25\alpha_1, 37\alpha_2, 49\alpha_3$ be three digit numbers,

then $\begin{vmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ 5 & 7 & 9 \\ 25\alpha_1 & 37\alpha_2 & 49\alpha_3 \end{vmatrix} =$

(A) $\alpha_1 + \alpha_2 + \alpha_3$ (B) $\alpha_1 - \alpha_2 + \alpha_3$

(C) 7 (D) 0

7. The roots of the equation

$$\begin{vmatrix} 3x^2 & x^2 + x\cos\theta + \cos^2\theta \\ x^2 + x\cos\theta + \cos^2\theta & 3\cos^2\theta \\ x^2 + x\sin\theta + \sin^2\theta & 1 + \frac{\sin 2\theta}{2} \end{vmatrix}$$

$$\begin{vmatrix} x^2 + x\sin\theta + \sin^2\theta \\ 1 + \frac{\sin 2\theta}{2} \\ 3\sin^2\theta \end{vmatrix} = 0$$ are

(A) $\sin\theta, \cos\theta$ (B) $\sin^2\theta, \cos^2\theta$

(C) $\sin\theta, \cos^2\theta$ (D) $\sin^2\theta, \cos\theta$

Q8. If a, b, c are sides of a triangle and

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 0$$, then

(A) ΔABC is an equilateral triangle

(B) ΔABC is a right angled isosceles triangle

(C) ΔABC is an isosceles triangle

(D) None of the above

Q9. If $y = \cos\theta + \sin\theta$ and

$$\begin{vmatrix} 1+y & 1-y & 1-y \\ 1-y & 1+y & 1-y \\ 1-y & 1-y & 1+y \end{vmatrix} = 0,$$

then the number of values of $\theta \in [0, 2\pi]$ is

(A) 1 (B) 2

(C) 3 (D) None of these

Q10. If $a^2 + b^2 + c^2 = -2$ and

$$f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix},$$

then f(x) is a polynomial of degree-

(A) 2 (B) 3

(C) 0 (D) 1

Q11. If $a+b+c=0$, then all the real solution of the equation $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$ is

($x \in R$), (a, b, $c \in R$)

(A) 0

(B) $\pm\frac{3}{2}(a^2+b^2+c^2)$

(C) $0, \pm\sqrt{\frac{3}{2}(a^2+b^2+c^2)}$

(D) $0, \pm\sqrt{a^2+b^2+c^2}$

Q12. The determinant $\begin{vmatrix} 4+x^2 & -6 & -2 \\ -6 & 9+x^2 & 3 \\ -2 & 3 & 1+x^2 \end{vmatrix}$ is not divisible by

(A) x (B) x^3

(C) $14+x^2$ (D) x^5

Q13. If 3^n is a factor of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ {}^{n}C_1 & {}^{n+3}C_1 & {}^{n+6}C_1 \\ {}^{n}C_2 & {}^{n+3}C_2 & {}^{n+6}C_2 \end{vmatrix}$, then the maximum value of n is

(A) 7 (B) 5 (C) 3 (D) 1

Q14. If $a_i^2 + b_i^2 + c_i^2 = 1,\ (i = 1,2,3)$ and $a_i a_j + b_i b_j + c_i c_j = 0,\ (i \neq j, i, j = 1,2,3)$ then the value of $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$ is -

(A) 0 (B) 1/2 (C) 1 (D) 2

Q15. An equilateral triangle has each of its sides of length 4 cm. If $(x_i, y_i)(i = 1,2,3)$ are its vertices then the value of $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2$ is

(A) 192 (B) 768 (C) 1024 (D)128

Q16. The system of equations

$x - y\cos\theta + z\cos 2\theta = 0$

$-x\cos\theta + y - z\cos\theta = 0$

$x\cos 2\theta - y\cos\theta + z = 0$

has non trivial solution for θ equals :

(A) $n\pi$ only, $n \in I$

(B) $n\pi + \frac{\pi}{4}$ only, $n \in I$

(C) $(2n-1)\frac{\pi}{2}$ only, $n \in I$

(D) all value of θ

Q17. If a, b, c are non zeros,then the system of equations $(\alpha+a)x + \alpha y + \alpha z = 0$, $\alpha x + (\alpha+b)y + \alpha z = 0$, $\alpha x + \alpha y + (\alpha+c)z = 0$ has a non-trivial solution if

(A) $\alpha^{-1} = -(a^{-1} + b^{-1} + c^{-1})$

(B) $\alpha^{-1} = a + b + c$

(C) $\alpha + a + b + c = 1$

(D) None of these

Q18. The system of equations

$\alpha x + y + z = \alpha - 1$

$x + \alpha y + z = \alpha - 1$

$x + y + \alpha z = \alpha - 1$

has no solution, If α is

(A) 1 (B) not –2

(C) either -2 or 1 (D) –2

Q19. If $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x \neq 0, y \neq 0$ then D is -

(A) Divisible by both x and y

(B) Divisible by x but not y

(C) Divisible by y but not x

(D) Divsible by neither x nor y

Q20. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that x = cy + bz, y = az + cx and z = bx + ay, then $a^2 + b^2 + c^2 + 2abc$ is equal to

(A) 2 (B) –1 (C) 0 (D) 1

Q21. If $\begin{vmatrix} 1+ax & 1+bx & 1+cx \\ 1+a_1x & 1+b_1x & 1+c_1x \\ 1+a_2x & 1+b_2x & 1+c_2x \end{vmatrix}$, $= A_0 + A_1x + A_2x^2 + A_3x^3$

then A_1 is equal to -

(A) abc (B) 0

(C) 1 (D) None of these

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $\Delta_r = \begin{vmatrix} 2^{r-1} & x & 2^n-1 \\ 2.3^{r-1} & y & 3^n-1 \\ 4.5^{r-1} & z & 5^n-1 \end{vmatrix}$, then $\sum_{r=1}^{n}\Delta_r$ is -

(A) independent of x
(B) independent of y
(C) independent of z
(D) independent of n

Q2. If

$$D_k\begin{vmatrix} 2^{k-1} & \frac{1}{k(k+1)} & \sin k\theta \\ x & y & z \\ 2^n-1 & \frac{n}{n+1} & \frac{\sin\left(\frac{n+1}{2}\right)\theta\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)} \end{vmatrix}$$

then $\sum_{k=1}^{n} D_k$ is

(A) equal to 0
(B) independent of n
(C) independent of θ
(D) independent of x, y and z

Q3. If f(x) and g(x) are functions such that $f(x+y) = f(x)\,g(y) + g(x)\,f(y)$, then

$\begin{vmatrix} f(\alpha) & g(\alpha) & f(\alpha+\theta) \\ f(\beta) & g(\beta) & f(\beta+\theta) \\ f(\gamma) & g(\gamma) & f(\gamma+\theta) \end{vmatrix}$ is independent of

(A) α (B) β (C) γ (D) θ

Q4. If $x \in N$ and ${}^{x}C_i$, ${}^{x^2}C_i$ and ${}^{x^3}C_i\,(i=1,2,3)$ are binomial coefficients,

then $12\begin{vmatrix} {}^{x}C_1 & {}^{x}C_2 & {}^{x}C_3 \\ {}^{x^2}C_1 & {}^{x^2}C_2 & {}^{x^2}C_3 \\ {}^{x^3}C_1 & {}^{x^3}C_2 & {}^{x^3}C_3 \end{vmatrix}$

is divisible by -

(A) x^3 (B) x^6
(C) x^9 (D) x^{12}

Q5. If

$$\Delta = \begin{vmatrix} 1/z & 1/z & -(x+y)/z^2 \\ -(y+z)/x^2 & 1/x & 1/x \\ -y(y+z)/x^2z & (x+2y+z)/xz & -y(x+y)/xz^2 \end{vmatrix}$$

then -

(A) Δ is independent of x
(B) Δ is independent of y
(C) Δ depends only on z
(D) $\Delta = 0$

Q6. If maximum and minimum values of the determinant

$$\begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$$

are α and β respectively then -

(A) $\alpha+\beta^{99} = 4$

(B) $a^3 - \beta^{17} = 26$

(C) $(\alpha^{2n} - \beta^{2n})$ is always an even integer for $n \in N$

(D) a triangle can be constructed having its sides as $\alpha-\beta, \alpha+\beta$ and $\alpha+3\beta$

Q7. If $a^2+b^2+c^2=1$, then

$$\begin{vmatrix} a^2+(b^2+c^2)\cos\phi & ab(1-\cos\phi) & ac(1-\cos\phi) \\ ba(1-\cos\phi) & b^2+(c^2+a^2)\cos\phi & bc(1-\cos\phi) \\ ca(1-\cos\phi) & cb(1-\cos\phi) & c^2+(a^2+b^2)\cos\phi \end{vmatrix}$$

is independent of

(A) a (B) b (C) c (D) ϕ

Q8. If p, q, r, s are in A.P. and

$$f(x)=\begin{vmatrix} p+\sin x & q+\sin x & p-r+\sin x \\ q+\sin x & r+\sin x & -1+\sin x \\ r+\sin x & s+\sin x & s-q+\sin x \end{vmatrix}$$

such that $\int_0^2 f(x)dx=-4$ then the common difference of the A.P. can be :

(A) -1 (B) $\frac{1}{2}$ (C) 1 (D) 2

Q9. If system of equation $px+qy+rz=0$, $qx+ry+pz=0$ & $rx+py+qz=0$ has a unique solution (here $p, q, r \in R$ & $p \neq q \neq r$), then system of equations

$(q+r)x+(r+p)y+(p+q)z=0$,

$(r+p)x+(p+q)y+(r+q)z=0$,

$(p+q)x+(q+r)y+(r+p)z=0$

(A) has only one solution

(B) has infinite many solution

(C) has no solution

(D) is consistent

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 10 to 11)

$$f(\theta)=\begin{vmatrix} 1 & \sin 2\theta & (\sin\theta+\cos\theta) \\ \sin 2\theta & 1 & (\sin\theta+\cos\theta) \\ (\sin\theta+\cos\theta) & (\sin\theta+\cos\theta) & 3 \end{vmatrix}$$

Q10. Period of $f(\theta)$ is -

(A) 2π (B) π

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Q11. Difference between greatest and least value of $f(\theta)$ is -

(A) 0 (B) 1

(C) 2 (D) 3

Paragraph # 2 (Ques. 12 to 14)

Consider the system of equations $2x+ay+6z=8$, $x+2y+bz=5$ and $x+y+3z=4$, then -

Q12. This system of equations will have infinite solutions if

(A) $b=3$

(B) $b=2$

(C) $a=3$

(D) $a=2$

Q13. The given system of equations will have no solution if

(A) $a=2, b\neq 3$

(B) $a\neq 2, b\neq 3$

(C) $a\neq 2, b=3$

(D) $a=2, b=3$

Q14. The given system of equations will have a unique solution if

(A) $a\neq 2, b\neq 3$

(B) $a\neq 2$

(C) $a=2, b=3$

(D) $a=2, b\neq 3$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column- II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15. $p(\theta)=\begin{vmatrix} -\sqrt{2} & \sin\theta & \cos\theta \\ 1 & \cos\theta & \sin\theta \\ -1 & \sin\theta & -\cos\theta \end{vmatrix}$, $q(\theta)=\sqrt{2}\begin{vmatrix} \sin 2\theta & 1 & 1 \\ \cos 2\theta & 2 & 3 \\ \cos 2\theta & 3 & 5 \end{vmatrix}$

$r(\theta)=\begin{vmatrix} \cos\theta & \sin\theta & \cos\theta \\ -\sin\theta & \cos\theta & \sin\theta \\ -\cos\theta & -\sin\theta & \cos\theta \end{vmatrix}$ and $s(\theta)=\begin{vmatrix} \sec^2\theta & 1 & 1 \\ \cos^2\theta & \cos^2\theta & \operatorname{cosec}^2\theta \\ 1 & \cos^2\theta & \cot^2\theta \end{vmatrix}$

Match the functions on the left with their range on the right.

(A) $p(\theta)$	(P) $[0, 1]$
(B) $q(\theta)$	(Q) $[0, 2\sqrt{2}]$
(C) $r(\theta)$	(R) $[-2, 2]$
(D) $s(\theta)$	(S) $[-\sqrt{5}-2, -\sqrt{5}+2]$

Q16. Let $\begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} = ax^5+bx^4+cx^3+dx^2+ex+f$

Match the entries from the following columns:

Column-I	**Column-II**
(A) The value of f is equal to	(P) 0
(B) The value of e is equal to	(Q) 1
(C) The value of a + c is equal to	(R) – 1
(D) The value of b + d is equal to	(S) 3

NUMERICAL TYPE

Q17. Find the value of the determinant

$$\begin{vmatrix} \frac{1}{z} & \frac{1}{z} & -\frac{(x+y)}{z^2} \\ -\frac{(y+z)}{x^2} & \frac{1}{x} & \frac{1}{x} \\ -\frac{y(y+z)}{x^2 z} & \frac{x+2y+z}{xz} & -\frac{y(x+y)}{xz^2} \end{vmatrix}$$

Q18. Let a, b, c are the solutions of the cubic $x^3 - 5x^2 + 3x - 1 = 0$, then find the value of thedeterminant $\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}$.

Q19. The number of positive integral solution of the equation $\begin{vmatrix} x^3+1 & x^2y & x^2z \\ xy^2 & y^3+1 & y^2z \\ xz^2 & yz^2 & z^3+1 \end{vmatrix} = 30$ is

Q20. If a, b, c, d > 0; $x \in R$ and $(a^2 + b^2 + c^2)x^2 - 2(ab + bc + cd)x + b^2 + c^2 + d^2 \le 0$. Then $\begin{vmatrix} 33 & 14 & \ln a \\ 65 & 27 & \ln b \\ 97 & 40 & \ln c \end{vmatrix}$ is equal to

Q21. If $a \ne p$, $b \ne q$, $c \ne r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$ then the value of $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c}$ is

Q22. If $a^2 + b^2 + c^2 + ab + bc + ca \le 0 \ \forall$ a, b, $c \in$ R, then find the value of the determinant

$$\begin{vmatrix} (a+b+2)^2 & a^2+b^2 & 1 \\ 1 & (b+c+2)^2 & b^2+c^2 \\ c^2+a^2 & 1 & (c+a+2)^2 \end{vmatrix}$$

Q23. $\Delta(x) =$

$$\begin{vmatrix} \sin 2x & e^x \sin x + x\cos x & \sin x + x^2 \cos x \\ \cos x + \sin x & e^x + x & 1 + x^2 \\ e^x \cos x & e^{2x} & e^x \end{vmatrix},$$

then find $\Delta(2)$.

Q24. Evaluate :

$$\begin{vmatrix} 2 & a+b+c+d & ab+cd \\ (a+b+c+d) & 2(a+b)(c+d) & ab(c+d)+cd(a+b) \\ ab+cd & ab(c+d)+cd(a+b) & 2abcd \end{vmatrix}$$

Q25. The coefficient of x in the expansion of

$$\begin{vmatrix} (1+x)^{22} & (1+x)^{44} & (1+x)^{66} \\ (1+x)^{33} & (1+x)^{66} & (1+x)^{99} \\ (1+x)^{44} & (1+x)^{88} & (1+x)^{144} \end{vmatrix}$$ is

SUBJECTIVE TYPE

Q26. Prove that

$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$$

$= (1 + a^2 + b^2)^3$.

Q27. Prove that

$$\begin{vmatrix} (\beta+\gamma-\alpha-\delta)^4 & (\beta+\gamma-\alpha-\delta)^2 & 1 \\ (\gamma+\alpha-\beta-\delta)^4 & (\gamma+\alpha-\beta-\delta)^2 & 1 \\ (\alpha+\beta-\gamma-\delta)^4 & (\alpha+\beta-\gamma-\delta)^2 & 1 \end{vmatrix}$$

$= -64(\alpha-\beta)(\alpha-\gamma)(\alpha-\delta)(\beta-\gamma)(\beta-\delta)(\gamma-\delta)$

Q28. If $a^2 + b^2 + c^2 = 1$ then show that the value of the determinant

$$\begin{vmatrix} a^2+(b^2+c^2)\cos\theta & ba(1-\cos\theta) & ca(1-\cos\theta) \\ ab(1-\cos\theta) & b^2+(c^2+a^2)\cos\theta & cb(1-\cos\theta) \\ ac(1-\cos\theta) & bc(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$$

simplifies to $\cos^2\theta$.

Q29. If $ax_1^2 + by_1^2 + cz_1^2 = ax_2^2 + by_2^2 + cz_2^2 = ax_3^2 + by_3^2 + cz_3^2 = d$ and $ax_2x_3 + by_2y_3 + cz_2z_3 = ax_3x_1 + by_3y_1 + cz_3z_1 = ax_1x_2 + by_1y_2 + cz_1z_2 = f$, then prove that

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = (d-f)\left[\frac{d+2f}{abc}\right]^{1/2} \quad (a, b, c \neq 0).$$

Q30. Let a, b, c, be real numbers with $a^2+b^2+c^2=1$, Show that the equation

$$\begin{vmatrix} ax-by-c & bx+ay & cx+a \\ bx+ay & -ax+by-c & cy+b \\ cx+a & cy+b & -ax-by+c \end{vmatrix} = 0$$

represents a straight line.

Q31. Show that, $\begin{vmatrix} a^2+\lambda & ab & ac \\ ab & b^2+\lambda & bc \\ ac & bc & c^2+\lambda \end{vmatrix}$ is divisible by λ^2 and find the other factor.

Q32. Solve the system of equations :

$$\left.\begin{aligned} z + ay + a^2x + a^3 &= 0 \\ z + by + b^2x + b^3 &= 0 \\ z + cy + c^2x + c^3 &= 0 \end{aligned}\right]$$

where $a \neq b \neq c$.

Q33. Solve the equations : $Kx+2y-2z=1$, $4x+2Ky-z=2$, $6x+6y+Kz=3$ considering specially the case when $K=2$.

Q34. Investigate for what values of λ, μ the simultaneous equations $x+y+z=6$; $x+2y+3z=10$ & $x+2y+\lambda z=\mu$ have :

(A) A unique solution

(B) An infinite number of solutions

(C) No solution

Answer Key

Ex - 1

1	2	3	4	5	6	7	8	9	10
D	C	A	B	C	D	D	D	B	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	B	C	C	C	A	D	B	A	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	C	A	B	B	B	A	B	D	C
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
D	C	B	B	B	B	D	C	B	D

Ex - 1A

1. B,C,D **2.** A,B,C,D **3.** A,C,D **4.** A,B,C **5.** A,D

6. B,C,D **7.** B,D **8.** A,D **9.** A **10.** B

11. B **12.** D **13.** A **14.** B **15.** B

16. C **17.** A **18.** A

19. (A) → (Q), (B) → (P), (C) → (S), (D) → (S) **20.** 4 **21.** 0

22. 0 **23.** 0 **24.** 250 **25.** 5 **26.** 0

27. (34, -34) **28.** (a,c, -b, a) **29.** $-(1-a)^2(1+a+a^2)^2 = -(1-a^3)^2$

32. No Solution **33.** $\lambda = 1, -5/3$ **34.** $\lambda = 1$, $\lambda = -2$

Ex - 2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	B	D	A	C	D	A	C	B	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	D	C	C	A	D	A	D	A	D
21									
B									

EX - 2A

1. A,B,C,D **2.** A,B,C,D **3.** A,B,C,D **4.** A,B,C **5.** A,B,D

6. A,B,C **7.** A,B,C **8.** A, C

9. A,D **10.** C **11.** B **12.** D **13.** C

14. A

15. (A) → (Q), (B) → (R), (C) → (R), (D) → (P)

16. (A) → (Q), (B) → (S), (C) → (R), (D) → (Q)

17. 0 **18.** 80 **19.** 3 **20.** 0 **21.** 2

22. 65 **23.** 0 **24.** 0 **25.** 0

32. $x = -(a + b + c)$, $y = ab + bc + ca$, $z = -abc$

33. If $K \neq 2$, $\frac{x}{2(K+6)} = \frac{y}{2K+3} = \frac{z}{6(K-2)} = \frac{1}{2\left(K^2+2K+15\right)}$

34. (A) $\lambda \neq 3$ (B) $\lambda = 3, \mu = 10$ (C) $\lambda = 3, \mu \neq 10$

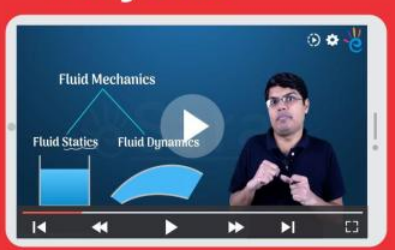
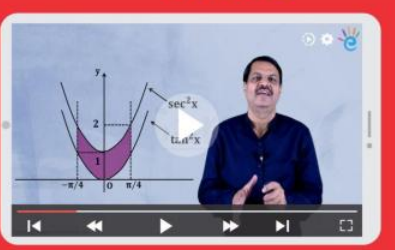
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